

All-Cause Mortality, Average of 1979-1983 and 1999-2000 Exposure

The coefficient and standard error for PM_{2.5} using the average of 1979-1983 and 1999-2000 PM data are estimated from the relative risk (1.06) and 95% confidence interval (1.02-1.11) associated with a change in annual mean exposure of 10.0 µg/m³ (Pope et al., 2002, Table 2).

G.1.1.10 Pope et al. (2015)

This cohort study evaluated the relationship between long-term exposure to ambient PM_{2.5} and risk of death from CVD and cardiometabolic disease, including effect modification of the relationship by pre-existing cardiometabolic risk factors, in the ACS Cancer Prevention Study II cohort (ages 30 and above). PM_{2.5} exposures were estimated at home addresses based on a land use regression and Bayesian maximum entropy (LUR-BME) interpolation model that incorporated data from ground-based monitors. Pope et al. utilized a Cox proportional hazards model controlling for individual-level covariates which included variables that characterized current and former smoking habits, exposure to second-hand cigarette smoke, workplace PM_{2.5} exposure in each subject's main lifetime occupation, self-reported exposure to dust and fumes in the workplace, marital status, level of education, body mass index, consumption of alcohol, and quartile ranges of dietary fat index and quartile ranges of a dietary vegetable/fruit/fiber index. Ecological covariates included median household income; percentage of people with <125% of poverty level income; percentage of unemployed individuals aged ≥16 years; percentage of adults with <12th grade education; and percentage of the population who were Black or Hispanic.

Mortality, Ischemic Heart Disease (ICD-10 code I20-I25)

In a single-pollutant model, the coefficient and standard error are estimated from the hazard ratio (1.14) and 95% confidence intervals (95% CI: 1.10-1.18) for a 10 µg/m³ increase in monthly PM_{2.5} exposure levels averaged from 1999-2004 (Pope, et al., 2015, Table 1. Cox model with individual-level plus ecological covariates; exposure based on LUR-BME).

G.1.1.11 Thurston et al. (2016)

Thurston et al. (2016) reevaluated the relationship between long-term exposure to ambient PM_{2.5} and mortality given the recent decline in U.S. ambient PM concentrations and differentiated risk for fossil fuel PM_{2.5} versus total PM_{2.5}. This cohort study analyzed all-cause, CVD, and respiratory mortality in the NIH-AARP cohort (ages 55-85) in 100 metropolitan areas in CA, GA, FL, LA, MI, NC, NJ, and PA. PM_{2.5} exposures were estimated at census tract centroids based on land use data and U.S. EPA AQS ground-based monitors. Participant exposure was estimated at census-tract of residence and included annual mean concentration in the year of mortality and 1-year lag average in a time-dependent model. Thurston et al. utilized a Cox proportional hazards model adjusted for race, education, marital status, BMI, alcohol consumption, smoking history,

and contextual variables such as median household income and percentage of population with less than a high school education.

Mortality, All-Cause

In a single-pollutant model, the coefficient and standard error are estimated from the hazard ratio (1.03) and 95% confidence intervals (95% CI: 1.00-1.05) for a 10 $\mu\text{g}/\text{m}^3$ increase in the annual mean $\text{PM}_{2.5}$ exposure level (Thurston, et al., 2016, Table 2. NIH-AARP cohort time independent Cox model $\text{PM}_{2.5}$ mortality hazard ratios (and 95th percentile CI) per 10 $\mu\text{g}/\text{m}^3$, by cause and cohort subgroup; cohort: ALL).

G.1.1.12 Turner et al. (2016)

See full study description under Turner et al. (2016) in Appendix E, Section E.1.2.

Mortality, Lung Cancer

In a single-pollutant model, the coefficient and standard error are estimated from the hazard ratio (1.09) and 95% confidence intervals (95% CI: 1.03-1.16) for a 10 $\mu\text{g}/\text{m}^3$ increase in the mean $\text{PM}_{2.5}$ exposure level from 1999-2004 (Turner, et al., 2016, Table E4: Lung cancer, fully-adjusted HR, LURBME $\text{PM}_{2.5}$).

G.1.1.13 Woodruff et al. (1997)

In a study of four million infants in 86 U.S. metropolitan areas conducted from 1989 to 1991, Woodruff et al. (1997) found a significant link between PM_{10} exposure in the first two months of an infant's life with the probability of dying between the ages of 28 days and 364 days. PM_{10} exposure was significant for all-cause mortality. PM_{10} was also significant for respiratory mortality in average birth-weight infants, but not low birth-weight infants.

Post-Neonatal Mortality

The coefficient and standard error are based on the odds ratio (1.04) and the 95% confidence interval (1.02-1.07) associated with a 10 $\mu\text{g}/\text{m}^3$ change in PM_{10} (Woodruff et al., 1997, Table 3).

G.1.1.14 Woodruff et al. (2006)

Studies suggest that airborne particulate matter (PM) may be associated with postneonatal infant mortality, particularly with respiratory causes and sudden infant death syndrome (SIDS). To further explore this issue, the authors examined the relationship between long-term exposure to fine PM air pollution and postneonatal infant mortality in California. They linked monitoring data for $\text{PM}_{2.5}$ to infants born in California in 1999 and 2000 using maternal addresses for mothers who lived within 5 miles of a $\text{PM}_{2.5}$ monitor. They matched each postneonatal infant death to four infants surviving to 1 year of age, by birth weight category and date of birth (within 2 weeks). For each matched set, they calculated exposure as the average $\text{PM}_{2.5}$ concentration over the period of life for the infant who died. They used conditional logistic regression to

estimate the odds of postneonatal all-cause, respiratory-related, SIDS, and external-cause (a control category) mortality by exposure to PM_{2.5}, controlling for the matched sets and maternal demographic factors. They matched 788 postneonatal infant deaths to 3,089 infant survivors, with 51 and 120 postneonatal deaths due to respiratory causes and SIDS, respectively. They found an adjusted odds ratio for a 10-microg/m³ increase in PM_{2.5} of 1.07 [95% confidence interval (CI), 0.93-1.24] for overall postneonatal mortality, 2.13 (95% CI, 1.12-4.05) for respiratory-related postneonatal mortality, 0.82 (95% CI, 0.55-1.23) for SIDS, and 0.83 (95% CI, 0.50-1.39) for external causes.

Post-Neonatal Mortality

The coefficient and standard error for PM_{2.5} are estimated from the relative risk (1.07) and the 95% confidence interval (0.93-1.24) associated with a change in annual mean exposure of 10.0 µg/m³ (Woodruff et al., 2006, p. 786).

G.1.1.15 Zanobetti et al. (2014)

Zanobetti et al. (2014) estimated the effect of short-term exposure to PM_{2.5} on all-cause mortality and also assessed the potential for pre-existing diseases (neurological disorders and diabetes) to modify the association between PM_{2.5} and mortality. This case-crossover analysis was a community-level assessment (community defined as the county or contiguous counties encompassing a city's population) of a Medicare cohort (age 65+) in 121 U.S. communities. PM_{2.5} exposures were estimated based on data from U.S. EPA AQS ground-based monitors. Daily PM_{2.5} data available for various monitors were averaged over the communities and participants were assigned a 2-day moving average (lag 0 and 1) based on community of residence. Zanobetti et al. (2014) first performed logistic regression at the community level, controlling for confounders such as the average temperature for the same and previous day. In a second stage of analysis, the community specific results were combined using multivariate meta-analysis techniques.

Short-term Mortality, All-Cause

In a single-pollutant model, the coefficient and standard error are estimated from the percent increase (0.64%) and 95% confidence intervals (95% CI: 0.42-0.85%) for a 10 µg/m³ increase in the 2-day average PM_{2.5} concentration (Zanobetti, et al., 2014, Table 2. Percent increase for 10 µg/m³ increase in the two days average PM_{2.5}: Combined across the 121 communities).

G.1.2 Chronic/Severe Illness

Table G-6 below summarizes the additional health impacts functions used to estimate the relationship between PM_{2.5} and chronic / severe health effects. We present a brief summary of each of the studies and any items that are unique to the study.

Table G-6. Additional Health Impact Functions for Particulate Matter and Chronic Illness

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Chronic Bronchitis	Abbey et al.	1995	SF, SD, South Coast Air Basin	27-99		Annual	0.013185	0.006796	Logistic	
Acute Myocardial Infarction, Nonfatal	Peters et al.	2001	Boston, MA	18-99		D24HourMean	0.024121	0.009285	Logistic	
Acute Myocardial Infarction, Nonfatal	Pope et al.	2006	Greater Salt Lake City, UT	0-99		D24HourMean	0.0048	0.0019	Logistic	Index MI and unstable angina
Acute Myocardial Infarction, Nonfatal	Sullivan et al.	2005	King County, WA	0-99		D24HourMean	0.0019	0.0022	Logistic	
Acute Myocardial Infarction, Nonfatal	Zanobetti and Schwartz	2006	Greater Boston, MA	0-99		D24HourMean	0.0053	0.0022	Logistic	Age range adjusted Admissions through ER visits only.
Acute Myocardial Infarction, Nonfatal	Zanobetti et al.	2009	26 U.S. Comm	0-99		D24HourMean	0.0022	0.0006	Log-linear	Age range adjusted. All Seasons.

G.1.2.1 Abbey et al. (1995b)

Abbey et al. (1995b) examined the relationship between estimated PM_{2.5} (annual mean from 1966 to 1977), PM₁₀ (annual mean from 1973 to 1977) and TSP (annual mean from 1973 to 1977) and the same chronic respiratory symptoms in a sample population of 1,868 Californian Seventh Day Adventists. The initial survey was conducted in 1977 and the final survey in 1987. To ensure a better estimate of exposure, the study participants had to have been living in the same area for an extended period of time. In single-pollutant models, there was a statistically significant PM_{2.5} relationship with development of chronic bronchitis, but not for AOD or asthma; PM₁₀ was significantly associated with chronic bronchitis and AOD; and TSP was significantly associated with all cases of all three chronic symptoms. Other pollutants were not examined.

Chronic Bronchitis

The estimated coefficient (0.0137) is presented for a one µg/m³ change in PM_{2.5} (Abbey et al., 1995b, Table 2). The standard error is calculated from the reported relative risk (1.81) and 95% confidence interval (0.98-3.25) for a 45 µg/m³ change in PM_{2.5}.

Incidence Rate: annual bronchitis incidence rate per person (Abbey et al., 1993, Table 3) = 0.00378